

Appendix S1

“Causal drivers of climate-mediated coral reef regime shifts”

Ecosphere

Suchinta Arif, Nicholas A. J. Graham, Shaun Wilson, M. Aaron MacNeil

S1 – Application of SCM

Applying SCM to determine causal relationships from observational data require 4 key steps outlined in Manuscript Figure 1. Here we detail each step that was taken to determine the causal drivers of coral-algal regime shifts following the 1998 bleaching event in Seychelles.

S1.1 - Step 1: Create DAG

We created our DAG by consolidating our domain knowledge with past and ongoing research, as well as the expert opinion of NAJG, SKW and MAM, who have extensive knowledge of Seychelles coral reefs in particular. We included relevant variables required to show how factors may influence climate-induced regime shifts across Seychelles coral reefs and included common causes of any pair of variables included in the DAG (Sprites et al. 2001, Glymour and Greenland, 2008). The justification of the causal links between each variable in our DAG is detailed in Table S1.

Table S1. Justification for the causal links in our coral reef DAG (Manuscript Figure 2).

Causal Assumptions	Rational
MPA -> Herbivorous Fish Biomass	It is widely accepted among coral reef scientists that MPAs generally lead to higher herbivorous fish biomass through the reduction of fishing pressure (e.g. Edwards et al. 2014; Soler et al. 2015). In Seychelles, MPAs have led to higher herbivorous biomass under stable conditions

	(Jennings et al. 1996), as well as after bleaching and the subsequent loss of corals (Graham et al. 2020).
Structural Complexity -> Herbivorous Fish Biomass	Structural complexity provides a wider range of habitat variability for reef fish species, which can increase herbivorous fish biomass (Graham and Nash, 2013; Rogers et al. 2014; Verges et al. 2011).
Structural Complexity -> Regime Shift	Structural complexity can play a critical role in reef recovery by providing niche space for coral settlement and survival (Connell et al. 1997; Victor 2008). It can also increase the abundance and diversity of reef fish through habitat availability and refuge (Verge's and Vanderklift, 2011; Graham and Nash 2013), which in turn can limit algal growth and regime shifts (Bellwood et al. 2004; Mumby and Harborne 2010). Structural complexity has played an important role in coral recovery in Guam (Colgan 1987) and the eastern Pacific (Guzman and Cortes 2007) and has previously been correlated with post-bleaching recovery in Seychelles (Graham et al. 2015).
MPA -> Branching Coral	MPAs had higher branching coral cover relative to fished sites across Seychelles coral reefs prior

	to the 1998 coral bleaching event (Graham et al. 2020). There is some uncertainty regarding why higher branching coral cover was found in MPAs pre-bleaching, and MPA placement in areas with higher branching coral may be a presumed factor. However, one of the marine reserves representing three of our MPA sites was chosen based on bird conservation and not coral cover (Graham et al. 2020). We suspect that MPAs may have more pre-disturbance branching coral relative to fished sites as branching corals are easily damaged by fishing gear and other recreational activities (McManus et al. 1997; Cros and McClanahan 2003; Strain et al. 2019), particularly trap fishery (Stevens 2020).
Branching Coral -> Regime Shift	Branching coral are generally more susceptible to bleaching and mortality from heat stress (Loya et al. 2001; Sheppard et al. 2002). Moreover, once dead, the structures provided by branching corals erode rapidly, impacting reef fish and other organisms (Sheppard et al. 2002; Graham and Nash 2013; Wilson et al. 2019). This in turn can leave reefs with higher pre-disturbance branching coral cover more vulnerable to regime shifts following a climate-induced bleaching event.

Wave Exposure -> Regime Shift	Wave exposure can favor macroalgae growth through increased exposure and uptake of inorganic carbon and nutrients (Larned and Atkinson, 1997; Hurd 2000). Wave action can also limit coral growth and larval settlement (Gove et al. 2015), as well as remove coral through colony dislodgement and abrasive damage (Madin and Connolly 2006). These factors are expected to increase the chances of regime shifting at wave-exposed sites. However, we note that higher flow rates have also been attributed to reduced bleaching susceptibility and faster recovery of corals (Nakamura and Woesik 2001; Nakamura et al. 2003), which in turn may limit coral-algal shifts following a climate-induced bleaching event.
Depth -> Regime Shift	Many threats, such as sediment input and fishing pressure, are less pronounced in deeper waters, making them less vulnerable to regime shifts (Bridge et al. 2013). Reduced light penetration and temperature may also decrease the likelihood of heat stress and subsequent coral bleaching (Slattery et al. 2011; Bridge et al. 2013). However, we note that light is also required for recovery of corals and on turbid reefs, coral may

	recovery more quickly in shallow water (Evans et al. 2020). Overall, we expect that depth will likely lower the changes of a regime shifting, which has previously been correlated with decreased likelihood of regime shifts in Seychelles (Graham et al. 2015).
Nutrient -> Regime Shift	Increased nutrient loads from human sources can increase macroalgae growth and abundance (Schaffelke and Klumpp 1997; Burkepile and Hay, 2006), limit coral growth rates (Koop et al. 2001), reproductive success and settlement (Harrison and Wallace 1990; Ward and Harrison 1997), as well as increase bleaching susceptibility of corals (Wooldrige 2009). Increased nutrient loading in combination with reduced herbivory has also been shown to make coral reefs less resilient against disturbances (Hughes et al. 2003; Burkepile and Hay 2006). Collectively, these factors are expected to increase the likelihood of regime shifting for reefs with higher nutrients, a correlation that has previously been shown in Seychelles (Graham et al. 2015).

Depth -> Nutrient	Nutrients tend to decrease with depth because shallow reefs are more exposed to nutrient run-off (Bridge et al. 2013).
Depth -> Branching Coral	Branching coral species inhabit shallow regions and are therefore expected to decrease with depth (e.g. Tamir et al. 2019).
Herbivore Fish Biomass -> Regime Shift	Herbivorous reef fish can limit coral-algal shifts through increased grazing pressure, which limits the growth of macroalgae and enhances coral recruitment through creating space for larval settlement (McCook et al. 2001; Bellwood et al. 2004; Mumby et al. 2006; Hughes 2007; Mumby et al. 2013).
Wave Exposure -> Branching Coral	Wave exposure influences branching coral distribution patterns and colony size, with higher wave exposure reducing branching coral cover (Done, 1983; Madin and Connolly, 2006).
Branching coral -> Structural Complexity	Branching coral is positively correlated with structural complexity (Graham and Nash, 2013). Branching coral may be particularly likely to contribute to fine-scale structural complexity on reefs (Chabanet et al. 1997).
Initial Macroalgae -> Regime Shift	Higher cover of pre-disturbance macroalgae may create a stronger basis for macroalgae expansion

	post-disturbance given that macroalgae species tend to have limited dispersal ability, with propagule settlement and recruitment remaining close to the source population (e.g., Kendrick and Walker, 1991; Kendrick and Walker 1995; Capdevila et al. 2018).
Herbivore Biomass -> Initial Macroalgae	Herbivore fish can decrease macroalgal cover through increased grazing pressure (McCook et al. 2001; Bellwood et al. 2004; Mumby et al. 2006; Hughes 2007; Mumby et al. 2013).
Wave Exposure -> Initial Macroalgae	Wave exposure can increase macroalgae cover through increased exposure and uptake of inorganic carbon and nutrients (Larned and Atkinson, 1997; Hurd 2000).
Nutrients -> Initial Macroalgae	Nutrients can increase macroalgae cover by stimulating macroalgal growth (Schaffelke and Klumpp 1997; Burkepile and Hay, 2006).

S1.2 - Step 2: Test DAG-Data Consistency:

Once a DAG has been created, it can be directly tested against the associated data, to ensure DAG-data consistency. If an initial DAG does not pass DAG-data consistency it must be tweaked until DAG-data consistency is ensured. Testing DAG-data consistency emerges from **d-separation** rules, which is a DAG-based rule for deciding whether a variable X is independent of

another variable Y given a set of variable(s), Z in a *path*. A path is a sequence of arrows and nodes connecting two variables of interest within a DAG.

d-separation (Pearl, 1988): A set of variables, Z, is said to block (or d-separate) a path from one variable to another if either

- (i) the path contains at least one arrow-emitting variable that is in Z, or
- (ii) the path contains at least one collider variable (variable with two incoming arrows) that is outside Z and does not cause any variables in Z. A collider variable is a variable that is influenced by two or more variables in a given path. Collider variables block the association between the variables that influence it.

There are 32 conditional independencies (based on d-separation rules) that are implied by our DAG (see below). These independencies can be tested against our observational data in order to ensure DAG-data consistency. In other words, our observational data should be consistent with these conditional independencies. For example, given our DAG, MPA should be independent of wave exposure (see 1 below), and MPA should also be independent of macroalgae when herbivore biomass, nutrients, and wave exposure are adjusted for (see 2 below). Collectively, all 32 conditional independencies implied by our DAG should be consistent with our observational data to provide support for our overall DAG structure.

1. $MPA \perp \text{Wave Exposure}$ (*meaning MPA is independent of wave exposure*)
2. $MPA \perp \text{Macroalgae} \mid \text{Herbivore Biomass, Nutrients, Wave Exposure}$ (*meaning MPA is independent of macroalgae given that herbivore biomass, nutrients, and wave exposure are adjusted for*)

3. MPA $\perp$ Macroalgae | Depth, Herbivore Biomass, Wave Exposure
4. MPA $\perp$ Depth
5. MPA $\perp$ Nutrients
6. MPA $\perp$ Structural Complexity | Branching Coral
7. MPA $\perp$ Regime Shift | Branching Coral, Depth, Herbivore Biomass, Structural Complexity, Wave Exposure
8. Herbivore Biomass $\perp$ Wave Exposure | Branching Coral, MPA
9. Herbivore Biomass $\perp$ Wave Exposure | MPA, Structural Complexity
10. Herbivore Biomass $\perp$ Depth | Branching Coral, MPA
11. Herbivore Biomass $\perp$ Depth | MPA, Structural Complexity
12. Herbivore Biomass $\perp$ Nutrients | Depth
13. Herbivore Biomass $\perp$ Nutrients | Branching Coral, MPA
14. Herbivore Biomass $\perp$ Nutrients | MPA, Structural Complexity
15. Herbivore Biomass $\perp$ Branching Coral | MPA, Structural Complexity
16. Wave Exposure $\perp$ Depth
17. Wave Exposure $\perp$ Nutrients
18. Wave Exposure $\perp$ Structural Complexity | Branching Coral
19. Macroalgae $\perp$ Depth | Branching Coral, MPA, Nutrients, Wave Exposure
20. Macroalgae $\perp$ Depth | MPA, Nutrients, Structural Complexity, Wave Exposure

21. Macroalgae $\perp$ Depth | Herbivore Biomass, Nutrients, Wave Exposure
22. Macroalgae $\perp$ Branching Coral | Depth, MPA, Structural Complexity, Wave Exposure
23. Macroalgae $\perp$ Branching Coral | MPA, Nutrients, Structural Complexity, Wave Exposure
24. Macroalgae $\perp$ Branching Coral | Depth, Herbivore Biomass, Wave Exposure
25. Macroalgae $\perp$ Branching Coral | Herbivore Biomass, Nutrients, Wave Exposure
26. Macroalgae $\perp$ Structural Complexity | Branching Coral, Herbivore Biomass, MPA
27. Macroalgae $\perp$ Structural Complexity | Depth, Herbivore Biomass, Wave Exposure
28. Macroalgae $\perp$ Structural Complexity | Herbivore Biomass, Nutrients, Wave Exposure
29. Depth $\perp$ Structural Complexity | Branching Coral
30. Nutrients $\perp$ Branching Coral | Depth
31. Nutrients $\perp$ Structural Complexity | Branching Coral
32. Nutrients $\perp$ Structural Complexity | Depth

We used the R package ‘dagitty’ (Textor et al. 2016) to test DAG-data consistency between our observational data and specified DAG. Dagitty uses a formal test of zero (partial) correlation for each identified independency based on d-separation rules. To mitigate problems around multiple testing (e.g. for complex DAGs with many testable implications), the p-values obtained are then corrected using the Holm-Bonferroni method (see Textor et al. 2016 for details). These tests assume linearity and multivariate normality, though other options are available (see Ankur et al. 2021). Our observational data was found to be consistent with all 32 conditional independencies implied by our DAG. Once a DAG is finalized (i.e. ensures DAG-data consistency), we can

employ the backdoor criterion to guide model selection required to determine the effect of a predictor variable on a response variable of interest. We note that since several DAGs can pass DAG-data consistency, it is critical to ensure that a finalized DAG can be justified (or set of DAGs) based on theory, instead of relying solely on DAG-data consistency.

S1.3 - Step 3: Apply of the Backdoor Criterion for Model Selection:

The backdoor criterion is used to identify a set of variables, Z , that when adjusted for, allows the causal effect of X on Y to be accurately estimated from observational data, given a DAG. The backdoor criterion (Pearl, 1993, 2009) states that a set of variables, Z , is sufficient for estimating the causal effect of X on Y under two conditions:

1. The variables in Z block all backdoor paths from X to Y . A *path* within a DAG is any sequence of arrows and nodes connecting two variables of interest, X and Y . A *backdoor path* is a path between X and Y with an arrow pointing into both X and Y . The backdoor paths are the paths that create bias by providing an indirect, non-causal path along which information can flow. To block a backdoor path from X to Y , X and Y must be d-separated by adjusting for Z (see requirements for d-separation above).
2. No element of Z is a descendant of (i.e. caused by) X .

The back-door criterion blocks all non-causal pathways (i.e. backdoor paths) between our predictor and response variable of interest, while leaving all causal paths open. Note that while causality flows in the direction of DAG arrows, information can flow in both directions, inducing various sources of potential bias. As such, the application of backdoor criterion eliminates common statistical biases that can otherwise plague observational studies, including

confounding, overcontrol, and collider bias. Confounding bias occurs when a common cause between predictor and response is not adjusted for; overcontrol bias occurs when an intermediate variable between predictor and response is adjusted for, blocking the causal association between predictor and response; collider bias occurs when a variable affected by both predictor and response is adjusted for, creating a spurious association between predictor and response (see Elwert 2014 for detail). Collectively, given a DAG, the backdoor criterion guides model selection required to determine causal relationships without biased spurious correlations, theoretically equivalent to a randomized control experiment (Pearl 2009). Below we apply the backdoor criterion to guide model selection for determining the causal effect of each of our predictor variables. We note that only the effect size of the predictor variable should be interpreted for its associated model, as additional covariates are there as required controls.

MPA, depth and wave exposure models:

To determine the effect of MPA on regime shift trajectory, there are no backdoor paths that need to be blocked. Similarly, the effect of depth and wave exposure on regime shift also do not have any backdoor paths that need to be blocked. Therefore, no covariate adjustments are required for all three models.

Nutrient model:

To determine the effect of nutrient on regime shift, there are two back-door paths that need to be blocked or d-separated:

1. Regime Shift <- Depth -> Nutrients
2. Regime Shift <- Branching Coral <- Depth -> Nutrients

Adjusting for depth will block both backdoor paths. This satisfies condition 1 of the back-door criterion; condition 2 is also satisfied because depth is not a descendant of nutrients (i.e. it is not affected by nutrients). Therefore, the effect of nutrients on regime shift can be quantified by adjusting for depth (i.e. adding depth as a covariate) in our statistical model.

Herbivore biomass model:

To determine the effect of herbivores fish biomass on regime shift, there are two backdoor paths that need to be blocked or d-separated:

1. Regime Shift <- Structural Complexity -> Herbivore Biomass
2. Regime Shift <- Structural Complexity <- Branching Coral <- MPA -> Herbivore Biomass
3. Regime Shift <- Branching Coral <- MPA -> Herbivorous Fish Biomass

Adjusting for structural complexity (blocking path 1 and path 2) and branching coral (blocking path 3) will allow us to estimate the causal effect of herbivorous fish biomass on regime shift.

To determine the effect branching coral on regime shift, there are five backdoor paths that must be blocked or d-separated:

1. Regime Shift <- Wave Exposure -> Branching Coral
2. Regime Shift <- Depth -> Branching Coral
3. Regime Shift <- Nutrients <- Depth -> Branching Coral
4. Regime Shift <- Herbivorous Fish Biomass <- MPA -> Branching Coral
5. Regime Shift <- Structural Complexity -> Herbivorous Fish Biomass <- MPA -> Branching Coral

Adjusting for wave exposure (blocking backdoor path 1), depth (blocking backdoor paths 2 and 3), and MPA (blocking backdoor path 4) will allow us to estimate the causal effect of branching coral on regime shift. We note that backdoor path 5 is already blocked because herbivorous fish biomass – a collider variable – has *not* been adjusted for.

Structural complexity model:

To determine the effect of structural complexity on regime shift, there are six backdoor paths that must be blocked or d-separated, including:

1. Regime Shift <- Branching Coral -> Structural Complexity
2. Regime Shift <- Wave -> Branching Coral -> Structural Complexity
3. Regime Shift <- Herbivorous Fish Biomass <- MPA -> Branching Coral -> Structural Complexity
4. Regime Shift <- Depth -> Branching Coral -> Structural Complexity
5. Regime Shift <- Nutrients -> Initial Macroalgae <- Depth -> Branching Coral -> Structural Complexity
6. Regime Shift <- Initial Macroalgae <- Wave -> Branching Coral -> Structural Complexity

All back-door paths between regime shift and structural complexity can be blocked or d-separated by adjusting for branching coral cover.

Initial macroalgae model:

Lastly, to determine the effect of initial macroalgae cover on regime shift, there are ten backdoor paths that must be blocked or d-separated:

1. Regime Shift <- Herbivorous Fish Biomass -> Initial Macroalgae
2. Regime Shift <- Structural Complexity -> Herbivorous Fish Biomass -> Initial Macroalgae
3. Regime Shift <- Structural Complexity <- Branching Coral Cover <- MPA -> Herbivorous Fish Biomass -> Initial Macroalgae
4. Regime Shift <- Herbivorous Fish Biomass <- MPA -> Branching Coral
5. Regime Shift <- Depth -> Initial Macroalgae
6. Regime Shift <- Nutrients <- Depth -> Initial Macroalgae
7. Regime Shift <- Branching Coral <- Depth -> Initial Macroalgae
8. Regime Shift <- Structural Complexity <- Branching Coral <- Depth -> Initial Macroalgae
9. Regime Shift <- Wave -> Branching Coral <- Depth -> Initial Macroalgae
10. Regime Shift <- Nutrients -> Initial Macroalgae
11. Regime Shift <- Wave -> Initial Macroalgae
12. Regime Shift <- Branching Coral <- Wave -> Initial Macroalgae

Adjusting for herbivorous fish biomass will block or d-separate backdoor paths 1-4, adjusting for depth will block or d-separate backdoor paths 5-9, adjusting for nutrients will block backdoor path 10 and adjusting for wave will block backdoor paths 11 and 12.

S1.4 - Step 4: Statistical Model:

The final step in SCM is to choose our statistical model, to estimate effect sizes. As DAGs are non-parametric, they make no assumptions about the distribution of variables (e.g. normal) or the functional form of effects (e.g. linear, nonlinear, stepwise), making them compatible with a wide range of statistical methods. Any non-linear relationships between variables must come from domain knowledge, or have a sound theoretical justification to avoid chasing noise in the observed data. DAGs are also compatible with both frequentist and Bayesian statistical approaches since they are used to guide model selection, and not the analysis itself. We note that statistical models developed under the SCM framework are still beholden to the same issues of sample size and measurement error in terms of the precision of resulting estimates; however, they will be more causally accurate. Here, we have employed the same point estimation method used by our past correlative analysis (Graham et al. 2015): a Bayesian logistic regression (see Manuscript for detail).

Specifically, we applied a Bayesian logistic regression analysis to each of our causal models, where the response variable was 0 for recovering sites, and 1 for regime shifting sites. We standardized our data by subtracting the mean of each variable and dividing by 2 standard deviations in order to assess relative effect sizes of our predictor variables (Gelman and Hill, 2007). We ran our models using the ‘rethinking’ package on R, using weakly informative prior. For logistic regression using standardized data, Gelmen et al. (2008) recommend a Cauchy(0,10) prior for B_0 and a student_t(1,0,2.5) priors for $B_{1...N}$ as weakly informed priors. Recently, others have recommended a student_t(ν ,0,2.5) for $B_{1...N}$ where ν is $3 < \nu < 7$ (Glosh et al. 2015). We have chosen ν of 4 (student_t(4,0,2.5)) but have compared our results using a ν of 1-7, which

all gave similar results. Our final Bayesian logistic regression models (one for each predictor variable) were:

$$Y_i \sim \text{Bernoulli}(p_i)$$

$$\text{MPA model: } \text{logit}(p_i) = \beta_0 + \beta_1 \text{MPA}$$

$$\text{Nutrient model: } \text{logit}(p_i) = \beta_0 + \beta_1 \text{Nutrient} + \beta_2 \text{Depth}$$

$$\text{Herbivore biomass model: } \text{logit}(p_i) = \beta_0 + \beta_1 \text{Herbivore biomass} + \beta_2 \text{MPA} + \beta_3 \text{Structural complexity}$$

$$\text{Branching coral model: } \text{logit}(p_i) = \beta_0 + \beta_1 \text{Branching coral} + \beta_2 \text{MPA} + \beta_3 \text{Depth} + \beta_4 \text{Wave}$$

$$\text{Depth model: } \text{logit}(p_i) = \beta_0 + \beta_1 \text{Depth}$$

$$\text{Structural complexity model: } \text{logit}(p_i) = \beta_0 + \beta_1 \text{Structural complexity} + \beta_2 \text{Branching coral}$$

$$\text{Wave model: } \text{logit}(p_i) = \beta_0 + \beta_1 \text{Wave}$$

$$\text{Initial macroalgae model: } \text{logit}(p_i)$$

$$= \beta_0 + \beta_1 \text{Macroalgae} + \beta_2 \text{Herbivore biomass} + \beta_3 \text{Depth} + \beta_4 \text{Nutrients} + \beta_5 \text{Wave}$$

Priors (for standardized data):

$$\beta_0 \sim \text{Cauchy}(0,10)$$

$$\beta_{1,\dots,N} \sim \text{Student}_t(4,0,2.5)$$

S2 – Causal vs. Correlative Models:

Table S2. Results from causal (backdoor criterion) vs correlative models (causal salad; no covariates).

Predictor Variable	Backdoor Criterion	Causal Salad	No Covariates
MPA	0.06 [-1.20, 1.39]	-1.57 [-5.02, 1.01]	NA
Nutrient	-1.61 [-3.65, 0.05]	-0.33 [-2.62, 1.81]	-1.30 [-3.44, 0.23]

Herbivore Biomass	1.76 [-0.44, 4.04]	0.08 [-1.75, 3.68]	0.90 [-0.65, 2.62]
Branching Coral	0.84 [-1.21, 3.09]	-0.04 [-2.68, 2.71]	1.41 [-0.04, 3.12]
Depth	-2.12 [-4.19, -0.49]	-4.76 [-9.67, -1.46]	NA
Structural Complexity	-3.77 [-7.33, -1.63]	-4.66 [-9.33, -1.77]	-3.81 [-7.06, 1.62]
Wave Exposure	1.02 [-0.45, 2.54]	1.83 [-0.89, 5.16]	NA
Initial Macroalgae	3.3 [0.40, 10.14]	1.44 [-1.67, 6.92]	4.44 [1.11, 10.24]

Table S2 contrasts causal estimates of our predictor variables from our causal models (where we used our DAG and the application of the backdoor criterion to guide model selection) with a causal salad model (where all variables were placed into one model and subsequently interpreted) and no covariate models (where additional variables were not adjusted for). “NA” is noted in instances where the backdoor models coincided with our no covariate model (i.e., no covariate adjustments were required by the backdoor criterion). The results show that causal estimates can vary greatly depending on model selection, highlighting the importance of applying DAGs to visualize causal structure and guide appropriate causal analysis.

S3 - Lack of Causal Inference Among Other Regime Shift Studies:

We conducted a literature review on observational studies investigating factors that cause coral reef regime shifts. A systematic literature search was conducted using the Web of Science database, with the search terms “coral reef + (phase OR regime) shift”, which identified 1,262 results. We read through each abstract and selected studies that (1) used observational data and (2) focused on a form of coral reef phase shift (e.g. coral reef to macroalgae phase shift; recovery

back to coral reef state). We excluded studies that looked at coral cover but did not consider this specifically under a phase shift context. The summary of our 14 selected studies is highlighted in Table S3.

The questions motivating each study are causal (e.g. factors influencing coral-macroalgal phase shifts, Graham et al, 2015), and causal language such as “effect”, “influence”, or “cause” is used in all but three studies (Table S3). However, these studies have not employed any causal inference methods (Table S3). We note that while we have emphasized the use of SCM, there are other approaches to causal inference that are available to researchers. These include quasi-experimental methods employed under the potential outcomes (PO) framework (Butsic et al. 2017; Larsen et al. 2019), as well as convergent cross mapping (CCM; Sugihara 2012) and granger causality (Detto et al. 2011), which can be used on time series data.

Our study stands apart from past observational work on coral reef regime shifts because we have employed the back-door criterion to guide covariate selection. In doing so, we have explicitly stated the causal structure of our system via our DAG and have adjusted for specific variables to answer specific causal questions. In contrast, while the studies listed under Table S3 may explain why a predictor variable is expected to influence the response, none (including our own) state their causal assumptions about how all relevant variables are causally connected. For example, Jouffray et al. 2019 used 20 anthropogenic and biophysical predictors thought to influence regime shifts without stating how variables may be causally connected. Without such consideration, issues of over-adjustment can lead to biased estimates between variables of interest due to processes such as overcontrol bias and collider bias (Pearl 2009; Elwert 2013). On the other hand, some studies (Mumby and Harborne, 2010; Arias-Gonzalez et al., 2017) used only one predictor variable, without considering the need (or lack thereof) for covariate

adjustment, which in turn can lead to confounding bias (Pearl 2009; Elwert 2013). Collectively, past observational studies on coral reef regime shifts have not employed causal inference methods and may therefore be prone to spurious and potentially misleading conclusions. In contrast, SCM offers a strong causal inference framework for coral reef ecologists interested in cause and effect relationships. It allows researchers to utilize their domain ecological knowledge to create DAGs and guide appropriate model selection for causal analysis.

Table S3. Summary of observational studies that look at factors associated with coral reef regime shifts. * denotes when causal language was used in the title.

Paper	Predictor Variables	Response Variable	Conclusion	Causal Language	Causal Inference
Graham et al. 2015	Herbivorous biomass, juvenile coral density, depth, initial structural complexity, carbon nitrogen ratio of sampled algae	Recovery of coral reefs	Post-bleaching coral reef recovery favored when reefs were structurally complex, in deep water, with high density of juvenile corals and herbivorous fish and low nutrient loads	Affect; influence	No
Jouffray et al. 2015	Various; E.g. grazer biomass, caper biomass, brewer biomass, large predator biomass, latitude, depth, reef zone, etc.	Reef regimes (calcifying regime; turf regime; Macroalgal/sand regime)	Various; E.g. macroalgal regime decreased with increase in grazer, scrapper, and browser biomass	Drivers*; influence; effect, impact, explain	No

Cheal et al. 2010	herbivore diversity, scraper/excavator abundance, grazer/detritivore abundance, algal browser abundance	Phase 1 (low macroalgae), Phase 2 (transitional), Phase 3 (established macroalgae)	Low fish herbivore diversity and low abundance of algal browsers and grazers/detritivores correlated with coral reef recovery	None	No
Aronson and Precht 2000	Year	Urchin density, fish count, percent cover macroalgae, percent cover CTB, percent cover hard corals, percent cover Halimeda spp (separate ANOVA for each variable)	Macroalgae cover correlated with reduced urchin herbivory	None	No
Kumagai et al. 2018	Sea surface temperature, surface current, estimated herbivorous fishes occurrence	Macroalgae-coral shift	Ocean current and herbivory drive macroalgae-to-coral shift under climate warming	Drive*, effect, influence, promote	No
Jouffray et al. 2019	20 anthropogenic and biophysical variables (e.g. effluent, habitat modification, non-commercial boat fishing)	Occurrence of 4 distinct regime shifts	Various: E.g. Regime 2 was best predicted by a strong positive relationship with maximum monthly climatological mean of wave power	Drivers*, effect, influence, impact	No

Johns et al. 2018	Habitat	Coral recovery (coral recruitment, recruit survival, juvenile coral persistence)	Negative impact of macroalgae on recruitment and recruit survival of corals maintain macroalgae state	Maintain*, effect, increase, influence, impact, create	No
Mumby and Harborne 2010	Marine reserve status	Rate of change of coral cover	Marine reserves enhance the recovery of corals	Enhance, effect, affect, higher, lower, increase	No
Cruz et al. 2018	Anthropogenic and environmental factors	Zoanthid and macroalgal phase shift indices (PSI)	Marginal reefs: correlation between coral-zoanthid shifts and local human impacts; coral-algal shifts and ports, urbanized surfaces, higher latitudes and shore proximity.	Effect; increase, decrease,	No
Bennett et al. 2015	Year	Fish trophic biomass, cover of benthic functional groups (separate ANOVA for each response variable)	Tropical herbivores maintain coral reef state	Maintain, increase, reduce, cause, impact, drive, lower	No
Ledlie et al. 2007	Year and monitoring site	Percent coral cover, percent macroalgae, changes in abundance and	Absence of macroalgal consumers and current dominance	None	No

		biomass of herbivorous fish (separate ANOVA for each response variable)	of macroalgae reduces a shift back to coral-dominated state		
Arias-Gonzalez et al. 2017	Year (associated with coral-algal phase shift)	Length, abundance, biomass for herbivorous fish	Coral-algal phase shift not driven by herbivorous fish	Driven*, effect, affect, precipitate	No
Huges, 1994	NA: interpretation based on graphs showing changes in coral cover, human population, number of Diadema, etc. with time		Overfishing, hurricane damage and disease have combined to cause coral-macroalgal shift	Cause; effect;	No
Lapointe 1997	NA: nutrient-related variables were recorded for sites experiencing phase shifts, which exceeded nutrient thresholds noted to sustain macroalgal blooms		Nutrient enrichment is a causal factor for coral-macroalgae phase shift in Jamaica and southeast Florida	Causal factor	No

Literature Cited:

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